in limits can the interval be lie [8, -9] like upper limit is positive na dlower limit is negative and limit can be in decimal of integration drrivation or function

Good question 👍 Let's clear this step by step.

1️⃣ Can an interval be written like [8, −9]?

Technically, **no** — not in standard mathematical notation.

In intervals, we always write:

$$[a, b]$$

where

- **a = lower limit**
- **b = upper limit**
- and **a ≤ b**

So since **−9 < 8**, the correct way to write it is:

$$[-9, 8]$$

Writing **[8, −9]** is considered incorrect because the lower limit must be smaller than the upper limit.

2️⃣ Can lower limit be negative and upper limit be positive?

✅ **Yes, absolutely.**

$$\int_{-9}^{8} f(x)\,dx$$

This is completely valid. Many integrals cross zero.

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3 What if you reverse limits in integration?

If you write:

$$\int_{8}^{-9} f(x)\,dx$$

This is valid mathematically, but:

$$\int_{8}^{-9} f(x)\,dx = -\int_{-9}^{8} f(x)\,dx$$

So reversing limits changes the **sign** of the integral.

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4 Can limits be decimals?

✅ Yes — limits can be:

- Integers → $\int_{1}^{5}$
- Decimals → $\int_{2.5}^{7.8}$
- Fractions → $\int_{\frac{1}{2}}^{\frac{7}{3}}$
- Irrational numbers → $\int_{0}^{\pi}$
- Even infinity → $\int_{1}^{\infty}$

All are valid.

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5 What about limits in derivatives?

For limits (in calculus):

$$\lim_{x \to 2.5} f(x)$$

The value approaching can also be decimal, negative, fraction — anything real.

🔥 Final Summary

- ❌ [8, −9] → Incorrect interval form
- ✅ Correct form: [-9, 8]
- ✅ Limits can be negative

- ✅ Limits can be decimal
- ✅ Reversing integration limits changes the sign

If you want, I can also show a small numerical example to make it clearer 😊